Wellness Studio OS — AI & Agent Architecture Document

Document 7 · Version 1.0 · 29 September 2026 Sources: Master PRD v1.0, Solution Architecture v1.0 (SAD), Module Specification (MS). Scope: AI and agent architecture only. PRD content, solution architecture, database design and API contracts are referenced, not restated.

Labelling convention. Items taken from the sources are stated plainly. Items marked **[Proposed]** are design decisions this document adds to fill gaps (for team/mentor confirmation). Items marked **[Open]** are unresolved in the sources.

Source alignments and gaps to note

- The PRD lists the AI provider as undecided (Appendix C); the MS names Gemini or GPT-4o as candidates. This document stays provider-agnostic behind the LLM Gateway (SAD AD12).
- The sources define WhatsApp as **outbound, template-based, n8n-orchestrated** in Phase 1 (mock mode). Inbound conversational AI on WhatsApp is not in any committed phase. Section 12 designs it as a **[Proposed]** Phase 2+ extension.
- The MS lists five Admin Assistant tools (T-01 to T-06, incl. member count and dashboard summary); the PRD commits to two curated queries plus Create Member. The extra read tools are kept **behind the same feature flag**; the two PRD queries plus Create Member are the P0 acceptance set.
- The MS Feedback Assistant restricts raw comments to Owner/Admin; the SAD stores feedback in the App DB referencing ERP session IDs. This document follows both (see 3.5).

1. AI Strategy

1.1 AI Vision

An assistive, governed intelligence layer on top of the studio's operational system: AI helps people ask, decide and act faster, while the ERP remains the single source of business truth and people remain in control of consequential decisions (PRD §2.1, §4.5).

1.2 AI Objectives

#	Objective	Measure (from PRD §14.6)
O1	Prove governed AI actions end to end via the Admin Assistant	≥85% queries correct; ≥95% confirmed actions correct first time; **zero false-success**
O2	Reduce admin effort and prospect response time	Median Enquiry response <5 s; read queries p95 <8 s (NFR-PRF-04)
O3	Improve discovery, occupancy, retention (assistive)	Recommendation acceptance rate tracked
O4	Convert unstructured feedback into insight	≥80% agreement with human reviewer on sample
O5	Be agent-ready without redesign	Agents added as consumers of existing tools, audit and approval store
O6	Keep model provider replaceable	Provider switch is configuration, not code (R-10)

1.3 AI Design Principles

1. **ERP is truth, AI assists.** Payments, memberships, bookings, attendance are deterministic; AI drafts, summarises, ranks, classifies.
2. **Governed capabilities only.** AI never touches the database or ERP directly (BR-AI-01); it calls the same domain services the UI calls.
3. **Inherit, never exceed.** Tool visibility and execution use the invoking user's role, permissions and tenant (BR-AI-02).
4. **Ground every answer.** Responses are generated only from tool results or approved knowledge; refuse rather than guess (BR-AI-04, BR-AI-08).
5. **Propose, then confirm.** Writes are two-phase with arguments frozen at proposal time (SAD AD09).
6. **Rules choose, LLM explains.** Rule engines select eligible candidates; the LLM only ranks/words within that set.

7. **Fail visibly.** Unknown or failed states are reported as such; success is never claimed unless returned by the platform (BR-AI-05).
8. **Everything audited.** One immutable record per model-initiated tool call (NFR-AUD-03).
9. **Model-agnostic, tenant-gated.** Gateway abstraction; features gated by tier and feature flag (BR-AI-11).

1.4 Human-in-the-Loop Philosophy

Autonomy is earned in stages. The stage a capability sits at determines what the human must do:

Level	Name	Human role	Used by
L0	Inform	Reads an answer	Read queries, recommendations, feedback summaries
L1	Propose → Confirm	Confirms a frozen, displayed action	Create Member (P0)
L2	Draft → Approve	Reviews/edits/approves each output before any effect	CRM, Operations, Marketing agents
L3	Bounded auto-execution	Sets policy and reviews after the fact	**[Open]** Not planned; only after Phase 4 evidence, per action class, reversible actions only

Rules that never relax: advertising publication and spend always need explicit approval (BR-AI-10); agents never message members without approval (MS §7.1); opt-outs are enforced by the platform, not the agent.

1.5 Responsible AI Guidelines

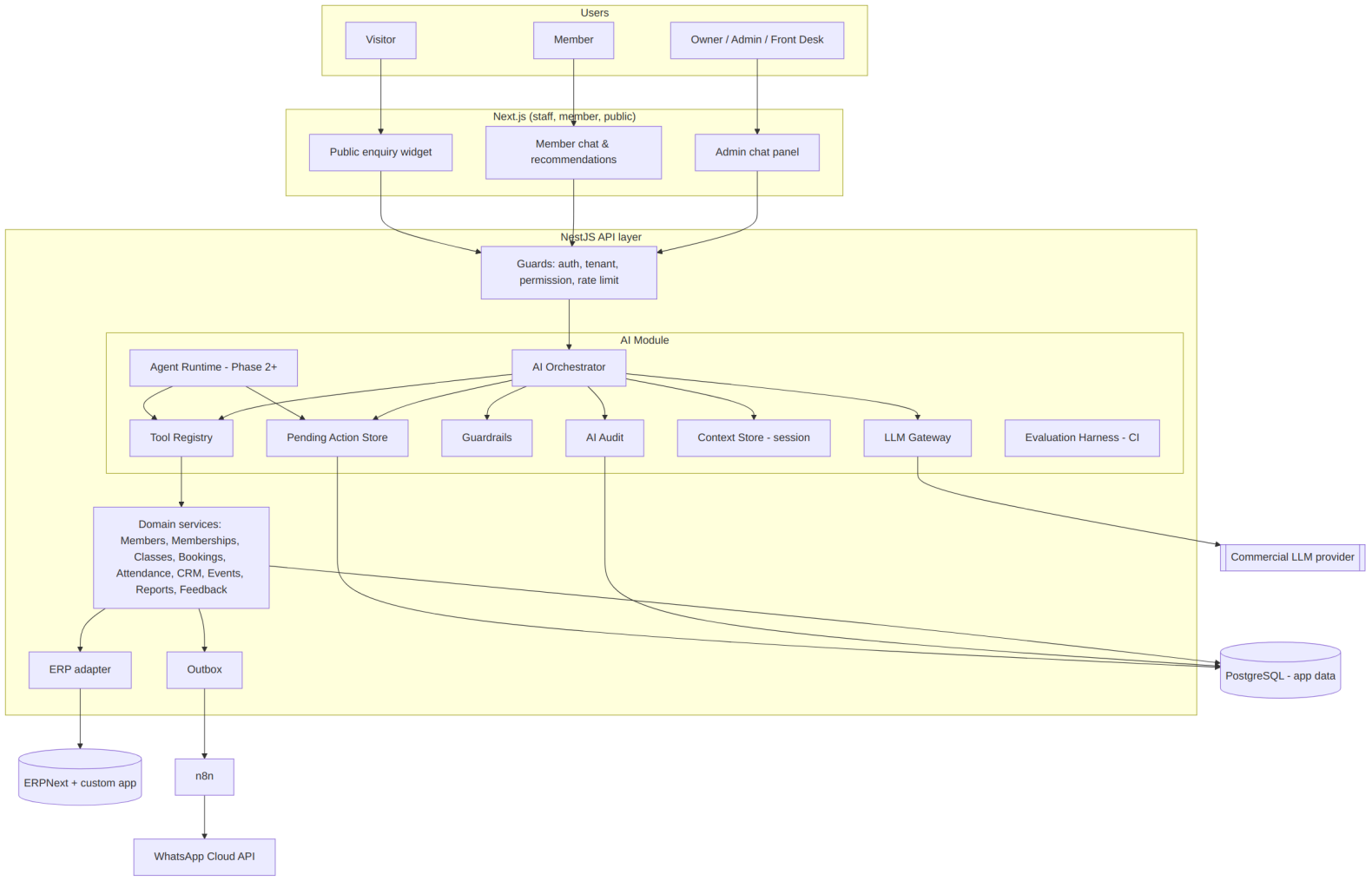
Area	Guideline
Transparency	Chat UI labels AI-generated content; confirmation cards show exact arguments
No health/medical/legal/financial advice	Hard scope refusal (BR-AI-08 in MS; BR-AI-09 in PRD "no medical claims"); goal/level are member-declared, never inferred
Data minimisation	PII minimised in prompts and logs (NFR-SEC-05); tools return only needed fields; health notes, payments, emergency contacts never reach recommendation flows
Fairness	Recommendations limited to entitlement; no upsell pressure; feedback never used to profile or act on individual members
Accountability	Named owners per assistant/agent; evaluation results recorded per release
Consent & compliance	Lead capture with consent notice; Indian data protection alignment and legal review pending (NFR-CMP-01, Appendix C)
Provider data privacy	Confirm what member data may be sent to the provider **[Open]** (PRD Appendix C)

2. AI Ecosystem Overview

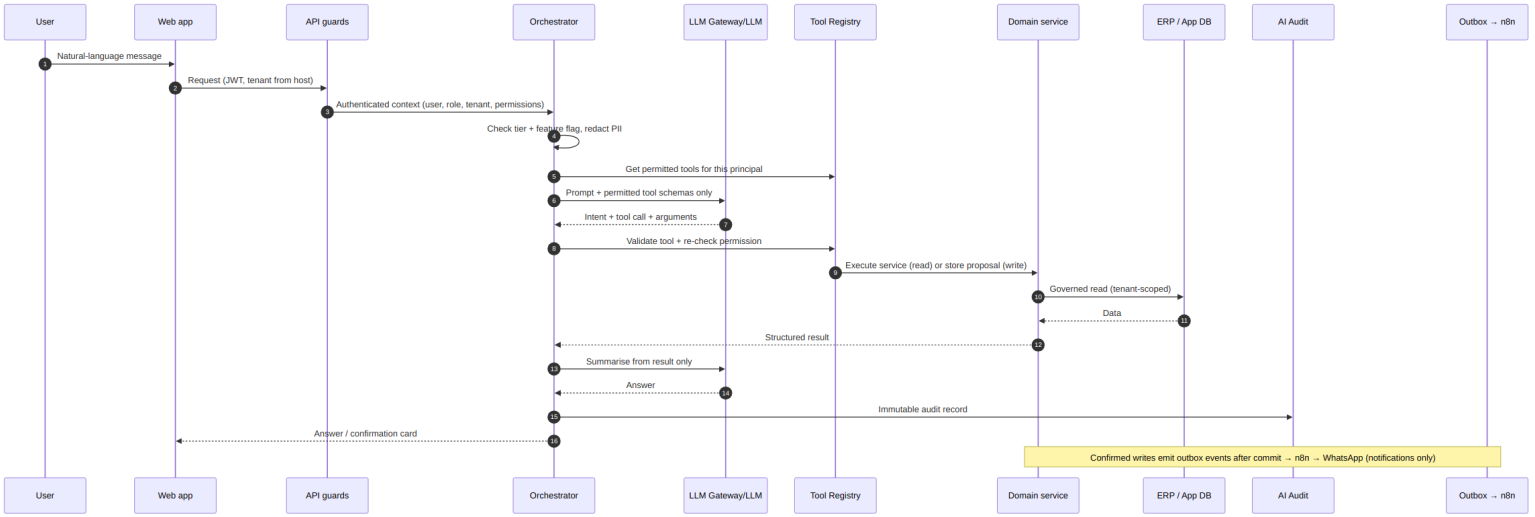
2.1 AI Architecture Overview

The AI layer is a **module inside the NestJS API layer in Phase 1** (SAD AD08), separable into its own worker/service later behind the same tool interface. Every AI request enters through a single Orchestrator; agents (Phase 2+) run as background workers on the same runtime.

2.2 AI Component Diagram



2.3 AI Data Flow Diagram



2.4 Interaction Summary

Actor	Interaction with AI
Users	Chat panels and cards; confirm/reject proposals
Application	Hosts the AI module; supplies identity, tenant, permission context
ERP	Indirect only: services read/write via the ERP adapter; AI has no ERP credentials
AI Services	Orchestrator, Gateway, Registry, Guardrails, Pending Actions, Audit
Automation layer	Consumes outbox events emitted by governed services; AI never calls n8n or WhatsApp directly in Phase 1

Actor	Interaction with AI
Future agents	Run on the same runtime and registry; propose into the approval queue; collaborate via outbox events, not agent-to-agent calls (SAD §9)

3. MVP AI Features

Common to all five: same Orchestrator, Gateway, Registry, Guardrails and Audit (Section 4); tier/flag gating server-side; degraded mode shows "assistant unavailable" while dashboards and UI remain unaffected (SAD §11).

3.1 AI Assistant (Admin) — P0 flagship

Item	Detail
Purpose	Natural-language operational queries and Create Member
Business value	Fewer clicks, faster decisions, proof of safe AI action
Inputs	Admin message; user, role, tenant; session context
Outputs	Concise answer; confirmation card; true action result with record ID or exact error; audit record
Data sources	Reporting, Members, Memberships, Attendance services (same as dashboard)
Tools (MS)	T-01 getExpiringMemberships (daysAhead=7), T-02 getLowAttendanceClasses (threshold 50%), T-03 extractCreateMemberFields (internal), T-04 createMember (write, confirm), T-05 getMemberCount, T-06 getDashboardSummary
Workflow	Message → intent → tool → grounded answer. Create Member: extract fields → ask for missing mandatory fields (never infer) → resolve relative dates (e.g. "Monday") in the service layer → store pending action → confirmation card → confirm → same `createMember` service as UI → report real result → audit
Failure handling	Validation/duplicate errors relayed verbatim; timeout → one bounded retry then "assistant unavailable"; permission denied → refusal with reason; ERP unavailable → "temporarily unavailable"; expired pending action → ask to restart; never claim success without a returned success
Security	Front Desk gets read tools only, if enabled; tool list filtered per user and re-checked at execution; PII not retained beyond session; scope refusal for health, tax, bulk messaging

3.2 AI Enquiry Assistant — P1 (narrow)

Item	Detail
Purpose	First-line answers for prospects; lead capture
Business value	Faster response, less staff load, higher lead capture
Inputs	Visitor question; tenant from host
Outputs	Grounded answer; lead (source = AI_ENQUIRY) after consent; handoff message
Data sources	Public class/event/plan data via public-scope services (T-07 getPublicEventList, T-08 getPublicClassSchedule, T-09 getActivePlansPublic); tenant-approved FAQ (App DB)
Workflow	Question → retrieve public data/FAQ → grounded answer → if interested collect name + phone (required), email, interest → consent notice → `captureLead` (T-10, same `createLead` as the manual form) → thank-you → outbox `lead.created` → WhatsApp confirmation. Else escalate to human
Failure handling	Insufficient source coverage or low confidence → escalate; CRM failure → apologise, offer manual contact form, no fake "lead saved"; duplicate lead flagged by CRM rules
Security	Restricted visitor principal: public-read tools + create-lead only; rate-limited and bot-protected; no member-private data; never creates Members; no invented prices/promises

3.3 AI Booking Assistant — P1 (narrow; cut first)

Item	Detail
Purpose	Shortlist eligible, available sessions and hand off to standard booking
Business value	Higher booking conversion, lower support load
Inputs	Member preference (type, day, time); member identity
Outputs	Ranked shortlist with reasons and deep links
Data sources	T-11 getEligibleSchedule, T-12 getAvailability, T-13 getMemberPlanEntitlement
Workflow	Preference → eligibility-filtered schedule → live availability → shortlist + links → member books in the standard flow (final validation in ERP)
Failure handling	No matches → say so and suggest nearby alternatives; availability changes before booking → standard flow returns authoritative result; tool failure → fall back to normal schedule page
Security	Member scope from session only; no booking tool in Phase 1 (`bookSession` is not registered); 100% of suggested sessions eligible and available at suggestion time

3.4 AI Recommendation Assistant — P1 (narrow; cut first)

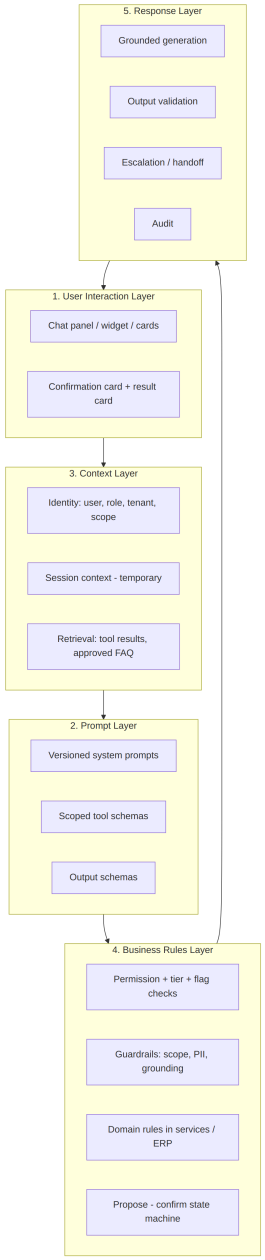
Item	Detail
Purpose	Explainable class suggestions
Business value	Discovery, occupancy, retention
Inputs	Declared goal and level; attendance counts by class type (30 days); plan entitlement
Outputs	Top 3–5 sessions with a short reason each
Data sources	T-14 getMemberGoalAndLevel, T-15 getMemberAttendanceSummary, T-16 getEligibleSessionsForRecommendation
Workflow	Read goal/level → attendance summary → fetch eligible upcoming sessions (7 days) → rank (goal alignment, variety, availability) → explain → link to standard booking
Failure handling	No goal declared → ask or show rules-only default; LLM unavailable → rules-only ranking with templated reasons
Security	Health notes, payments, emergency contact never read; no health claims; feedback comments never used to recommend to others

3.5 AI Feedback Assistant — P1 (narrow)

Item	Detail
Purpose	Turn free-text comments into structured insight
Business value	Faster detection of recurring issues and positive themes
Inputs	Optional comment with session reference; optional rating
Outputs	Sentiment (Positive/Neutral/Negative), topics (Trainer, Class Content, Music, Timing, Duration, Facility), one-sentence theme; per-session and weekly summaries
Data sources	Feedback records (App DB, referencing ERP session IDs)
Workflow	Submit → save raw comment first → async classification job (T-17) → store classification separately (T-18) → admin/trainer aggregate view (T-19)
Failure handling	Classification failure leaves raw comment intact with status "unclassified" and a retry; malformed model output rejected by schema validation
Security	Raw text readable only by Owner/Admin; Trainers see aggregates only; feedback is optional; advisory only, no automatic action toward the member

4. AI Assistant Architecture

4.1 Layered Architecture



Layer	Responsibilities	Key rules
User Interaction	Render chat, confirmation cards, real results and errors; deliver escalation messages	UI hides disallowed features, server enforces; every state has loading/error/denied views
Prompt	Versioned prompt templates in the repo; only permitted tool schemas injected; structured output schemas	Prompts contain policy, not data of other tenants; prompt versions recorded in audit
Context	Identity (from JWT, never from user text), ephemeral conversation state, retrieved data	Context never becomes business data except through a governed service (BR-AI-07)
Business Rules	Tier/flag gate, permission re-check at execution, guardrails, propose→confirm, rate limits	Deterministic rules in services/ERP stay authoritative; LLM cannot override them
Response	Generate from tool results only; validate; choose answer / clarify / refuse / escalate; write audit	No numbers or facts that did not come from a tool

4.2 Query Processing

1. Authenticate; bind tenant from host; load role and permissions.
2. Check tenant tier and feature flag; reject if disabled (server-enforced).
3. Redact PII from the user message where not required for the task.
4. Build permitted tool set (user × role × tenant × tier × flag).

5. LLM returns intent and tool call, or a clarifying question, or a refusal.
6. Orchestrator validates tool name, argument schema and permission again, then dispatches.

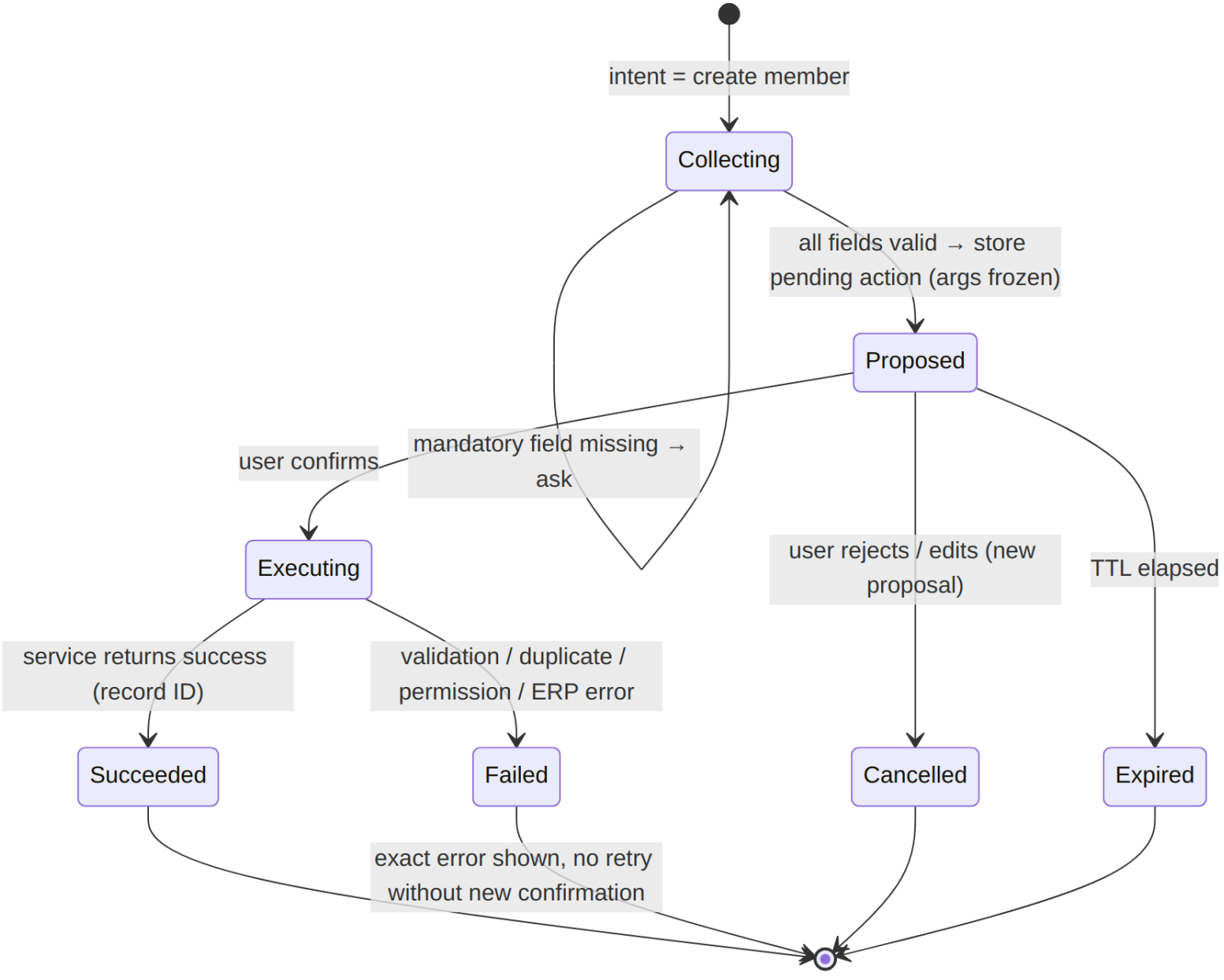
4.3 Context Retrieval

Two retrieval modes: **live tool retrieval** (structured business data from domain services; never cached for eligibility/availability) and **approved-knowledge retrieval** (Section 5–6; Enquiry Assistant). All retrieval is tenant-filtered; results are minimised to needed fields before returning to the model.

4.4 Response Generation

- Second LLM pass summarises **only** the returned data (T-01 result → "Which memberships expire this week?").
- Figures must equal dashboard figures (same Reporting service; B2B-DSH-01, NFR-USB-04). The evaluation harness asserts equality.
- Writes: the response is a confirmation summary from **stored** arguments; after confirmation, the response is the service's real result.

4.5 Create Member State Machine



4.6 Escalation to Human Staff

Trigger	Behaviour	Destination
Out-of-scope or unsupported request (Admin)	Scope message naming what the assistant can do	User (no staff)

Trigger	Behaviour	Destination
Enquiry: source coverage/confidence insufficient, pricing dispute, complaint, member-specific account issue	Handoff message; capture contact with consent; create lead/task	CRM lead (staff follow-up)
Repeated failure (LLM/tool)	"Assistant unavailable"; UI fallback link	Standard UI
Sensitive topic (health, legal, payment dispute)	Refuse and redirect to staff	Staff contact info configured by tenant
Explicit "talk to a person"	Immediate handoff	Staff via lead/task **[Proposed: admin task]**

5. Knowledge Base Architecture

5.1 Principle

Knowledge is **retrieved, not trained**: no model training and no large dataset (PRD A-06). Business truth is fetched live through governed services; only slow-changing text (FAQ, policies) is stored as knowledge.

5.2 Knowledge Sources

Source	Content	Owner / system	Consumer	Freshness
ERP knowledge	Classes, sessions, trainers, capacity, events, plans (public/permitted views)	ERP via services	Enquiry, Booking, Recommendation, Admin	Live (never cached for availability/eligibility)
Membership knowledge	Active plans, price, validity, allowed services, booking limits; member's own entitlement	ERP via Memberships service	Enquiry (public plans), Booking/Recommendation (own entitlement)	Live
Booking knowledge	Availability, waitlist counts, cancellation cutoff (config)	ERP via Bookings service; Config service	Booking Assistant, Enquiry	Live
Studio policies	Cancellation window, cutoff, opt-in/consent notice, house rules	Config service (App DB) + approved policy text	Enquiry, Booking	On change
FAQ	Tenant-approved Q&A (timings, what to bring, trial info, location)	App DB, authored in Configuration Center	Enquiry	On approval
Feedback corpus	Comments + classifications	App DB	Feedback Assistant, admin views	Async

5.3 Knowledge Storage

- **Structured knowledge:** stays in ERP/App DB; accessed only via tools. No copy in a vector index.
- **Unstructured knowledge (FAQ, policy text):** App DB table of tenant-scoped knowledge entries: tenant, title, body, category, status (Draft / Approved / Retired), version, approver, updated-at. **[Proposed]** (schema belongs to the Database Design document).
- Only **Approved** entries are retrievable (BR-AI-08 in PRD: approved knowledge only).

5.4 Knowledge Updates

1. Owner/Admin edits an entry in the Configuration Center → Draft.
2. Approval (Owner) → Approved; audit record written.
3. Re-index job for that tenant only (Phase 2 vector path); cache invalidated.
4. Retiring an entry removes it from retrieval immediately.
5. Seed FAQ template per vertical at onboarding; the tenant edits from there (configure, don't hard-code).
6. Structured data needs no update process because it is read live.

5.5 Retrieval Strategy

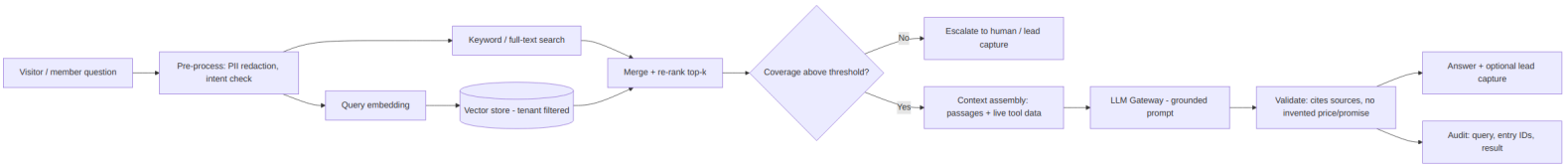
Phase	Strategy
Phase 1 (P1 narrow)	Small approved FAQ per tenant passed as context or matched by PostgreSQL full-text/keyword search; structured facts via tools. Tenant-filtered; entry IDs logged in audit
Phase 2 **[Proposed]**	Semantic retrieval over the approved FAQ/policy corpus (Section 6) with hybrid keyword + vector ranking, still tenant-filtered
Any phase	Answer must cite a tool result or an approved entry; no coverage → escalate (BR-AI-08)

6. RAG Architecture

6.1 Applicability

RAG is **applicable only to unstructured, approved tenant content** (FAQ, policies, later trainer-approved notes). Operational data uses tool calling, which is more accurate and permission-aware than embedding it. Phase 1 has a small corpus, so full retrieval infrastructure is optional; the design below is the Phase 2 target and the Phase 1 keyword path uses the same interfaces.

6.2 Retrieval Flow



6.3 Context Retrieval

- Filter first by **tenant and status = Approved**, then by audience (public vs member).
- Top-k small (e.g. 3–5) passages **[Proposed default]**; combine with live tool data for prices, schedule and availability, so prices are never answered from stale embedded text.
- Passages carry entry ID and version for audit and citation.

6.4 Embedding Strategy [Proposed]

- Chunk by FAQ entry (one Q&A = one chunk); policies chunked by section.
- Embedding model accessed through the LLM Gateway (provider-agnostic; embedding dimension recorded with the index version).
- Re-embed only changed entries on approval; index version tag enables rollback and model swap.
- No PII, member data or feedback is embedded.

6.5 Vector Store Strategy [Proposed]

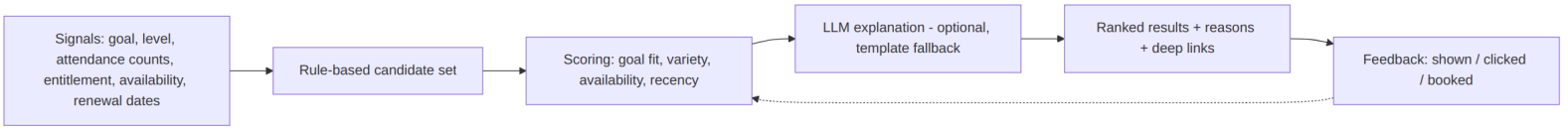
Aspect	Decision
Store	PostgreSQL vector extension in the existing App DB (no new infrastructure for a small team); revisit a dedicated store only on scale
Isolation	Tenant column + row-level security; **no cross-tenant index** (SAD §6.1); queries always tenant-filtered
Scale trigger	Corpus/latency growth → partition by tenant or extract to a dedicated store behind the same retrieval interface
Retention	Follows entry lifecycle; retired entries deleted from the index

6.6 Response Generation Flow

Assemble grounded prompt (system policy + passages + tool data) → LLM generates → validator checks the answer uses only supplied material, contains no medical advice and no invented price/promise → deliver, or fall back to escalation. Low-confidence or no-coverage results never produce a guess.

7. AI Recommendation Engine

Architecture: *Candidate generation (rules) → scoring (rules/statistics) → explanation (LLM, optional) → presentation*. Rules guarantee eligibility and availability; the LLM cannot add candidates.



7.1 Membership Recommendations

Item	Detail
Inputs	Member's plan, entitlement gaps (e.g. wants Yoga but plan excludes it), usage vs booking limit, expiry date, active plan catalogue
Logic	Rule: if a member repeatedly attempts/asks about excluded services or exhausts limits, surface a plan that includes them; renew prompt ≤7 days to expiry
Outputs	"Consider plan X because…" card; never pressure or auto-purchase
Future	Phase 2 upgrade/downgrade and freeze flows; Phase 3 propensity scoring; goal-to-plan assistant (F)

7.2 Class Recommendations (P1 narrow, MVP)

Item	Detail
Inputs	Declared goal/level, 30-day attendance by class type, entitlement, live availability
Logic	MS §6.5: eligible upcoming sessions (7 days) ranked by goal alignment, variety, availability; top 3–5
Outputs	Ranked sessions, one reason each, link to standard booking
Future	Phase 2 history-personalised ranking, time-of-day preference, acceptance-rate learning; goal-to-plan (F)

7.3 Event Recommendations

Item	Detail
Inputs	Event calendar, audience type (members-only vs public), member's class-type history, registration window and seats
Logic	Show only events the member is eligible for (members-only hidden from non-members); match event category to attended class types; prefer events with seats and open registration
Outputs	Short list with reason; link to registration/payment flow
Future	Phase 2 reminder preference and event waitlist; Phase 3 Marketing Agent uses the same logic for campaign segments

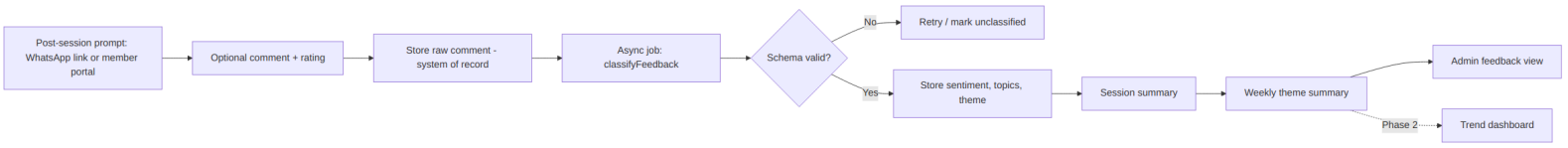
7.4 Retention Recommendations

Item	Detail
Inputs	Days since last attendance, attendance trend, expiry proximity, renewal history, booking cancellations
Logic	Phase 1: none automated (inactive nudges are Phase 2, 3 and 7 days). Phase 2: rule-based **at-risk lists** (e.g. no attendance in N days + expiring soon) for staff. Phase 3: forecast churn once data matures
Outputs	Admin-facing at-risk list with suggested action; drafts for approval (via CRM/Marketing agents)
Future	Studio Insights Agent churn forecasts; agent-recommended re-engagement (P3)

Measurement: acceptance rate (suggested → booked/registered) is logged per recommendation ID from Phase 1 to support Phase 2 personalisation (PRD §14.6).

8. AI Feedback Analysis

8.1 Workflow



Stage	Detail
Feedback collection	Optional, post-session, tied to a session reference; via member portal (P1) and an optional feedback link in the post-class WhatsApp message (P2). Never compulsory
Sentiment analysis	LLM classification into Positive/Neutral/Negative with constrained output schema; low-confidence flagged for human review; target ≥80% agreement on sampled set
Trend detection	Phase 1: weekly theme summary per class type. Phase 2: trend dashboard (topic frequency over time, sentiment shift by trainer/class/time slot). Phase 3: anomaly alerts through the Studio Insights Agent
Actionable insights	Aggregate-only outputs to Admin/Trainer (e.g. "Timing mentioned negatively in 6 of 20 Thursday evening comments"); suggested follow-ups are advisory and routed to staff, never sent to members automatically

Outputs: per-comment classification (Owner/Admin), per-session summary (Owner/Admin/Trainer aggregate), weekly theme summary. Raw comment is always retained as the record.

9. Future Agent Framework

Common agent contract (all agents): service principal scoped to one tenant and one domain toolset; user permissions are a ceiling when acting for a user; tools only from the registry; every tool call audited; write actions are proposals in the approval queue; rate-limited per tenant; prefer reversible recommendations (MS §7.1); no autonomous messaging.

9.1 CRM Agent (Phase 2)

Item	Detail
Goal	Reduce lead leakage; lift conversion
Responsibilities	Qualify/score leads; recommend next action; draft follow-ups; flag cold leads; track conversion
Inputs	Leads, sources, follow-up history, conversion events
Outputs	Lead score, next-action recommendation, draft messages, conversion insight
Triggers	`lead.created`; daily stale-lead scan; on-demand admin trigger
Tools (MS)	getLeadsWithStatus, scoreLead, draftFollowUpMessage, flagColdLead, proposeStatusChange
Approval	Required before any message is sent; status changes and cold flags are one-click approve/reject; scores advisory

9.2 Operations Agent (Phase 2)

Item	Detail
Goal	Continuous operational monitoring
Responsibilities	Monitor capacity, attendance, bookings, expiries; alert on low attendance/overbooking; propose schedule adjustments
Inputs	Classes, bookings, attendance, memberships
Outputs	Alerts, exception tasks, schedule suggestions
Triggers	Scheduled scans; occupancy threshold breach; repeated waitlist overflow; `booking.cancelled` patterns
Tools (MS)	getSessionOccupancyHistory, getAttendanceTrendByClassType, detectLowOccupancySessions, detectOversubscribedPatterns, proposeSessionCancellation, proposeCapacityIncrease, proposeScheduleChange
Approval	Alerts: none. Session cancellation, capacity change, any member-facing action: approval required; ERP still enforces rules

9.3 Marketing Agent (Phase 3)

Item	Detail
Goal	Retention, re-engagement, promotion
Responsibilities	Segment audiences; draft WhatsApp/email campaigns; submit for approval; hand approved sends to n8n; summarise performance
Inputs	Member segments, renewal/activity data, event calendar, consent/opt-in flags
Outputs	Draft campaigns, audience definitions, performance summaries
Triggers	Admin request; renewal/inactivity segments; upcoming event calendar
Approval	**Mandatory before every send**; opt-outs and consent enforced by the platform, not the agent; approved templates only

9.4 Advertising Agent (Phase 3)

Item	Detail
Goal	Efficient acquisition
Responsibilities	Draft ad copy; suggest audiences and landing pages; analyse campaign performance
Inputs	Lead source and conversion data, offers, event calendar
Outputs	Ad drafts, audience suggestions, optimisation recommendations
Triggers	Admin request; new event/offer; periodic performance review
Approval	**Mandatory before any publication or spend** (BR-AI-10); the agent holds no platform credentials with publish rights

9.5 Growth Optimization Agent (Phase 3)

Item	Detail
Goal	Revenue and utilisation
Responsibilities	Analyse revenue, occupancy, conversion; recommend pricing and schedule changes
Inputs	Revenue, occupancy, conversion, plan data
Outputs	Ranked, explained recommendations
Triggers	Monthly cycle; owner request
Approval	Advisory only; the owner acts through the normal UI

9.6 Studio Insights Agent (Phase 3)

Item	Detail
Goal	Proactive intelligence and forecasts
Responsibilities	Periodic briefings; anomaly detection; driver explanation; churn/demand forecasts when data matures
Inputs	Cross-module operational and financial data (role-scoped to Owner)
Outputs	Insight briefings, anomaly alerts, forecasts
Triggers	Weekly/monthly schedule; anomaly threshold
Approval	Read-only; briefings delivered without approval; any derived action goes through other agents' approval queues

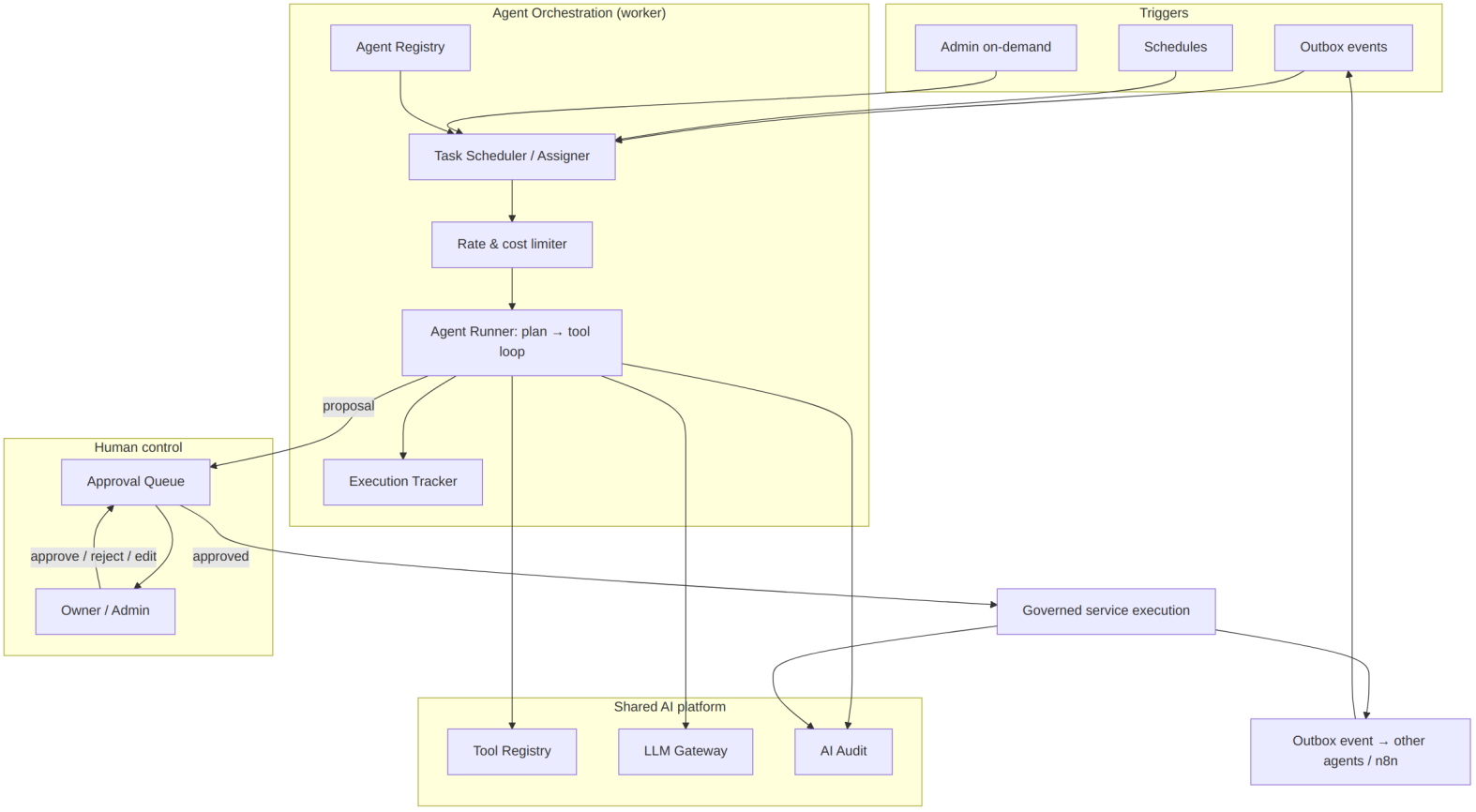
9.7 Session Summary Agent (Future)

Item	Detail
Goal	Readable progress and session summaries
Responsibilities	Member progress recaps; trainer/admin session summaries
Inputs	Attendance, session data, approved trainer notes
Outputs	Member-facing and staff-facing summaries
Triggers	Post-session or periodic job

Item	Detail
Approval	Trainer/admin review before member-facing delivery (recommended default); staff-facing summaries read-only

10. Agent Orchestration Architecture

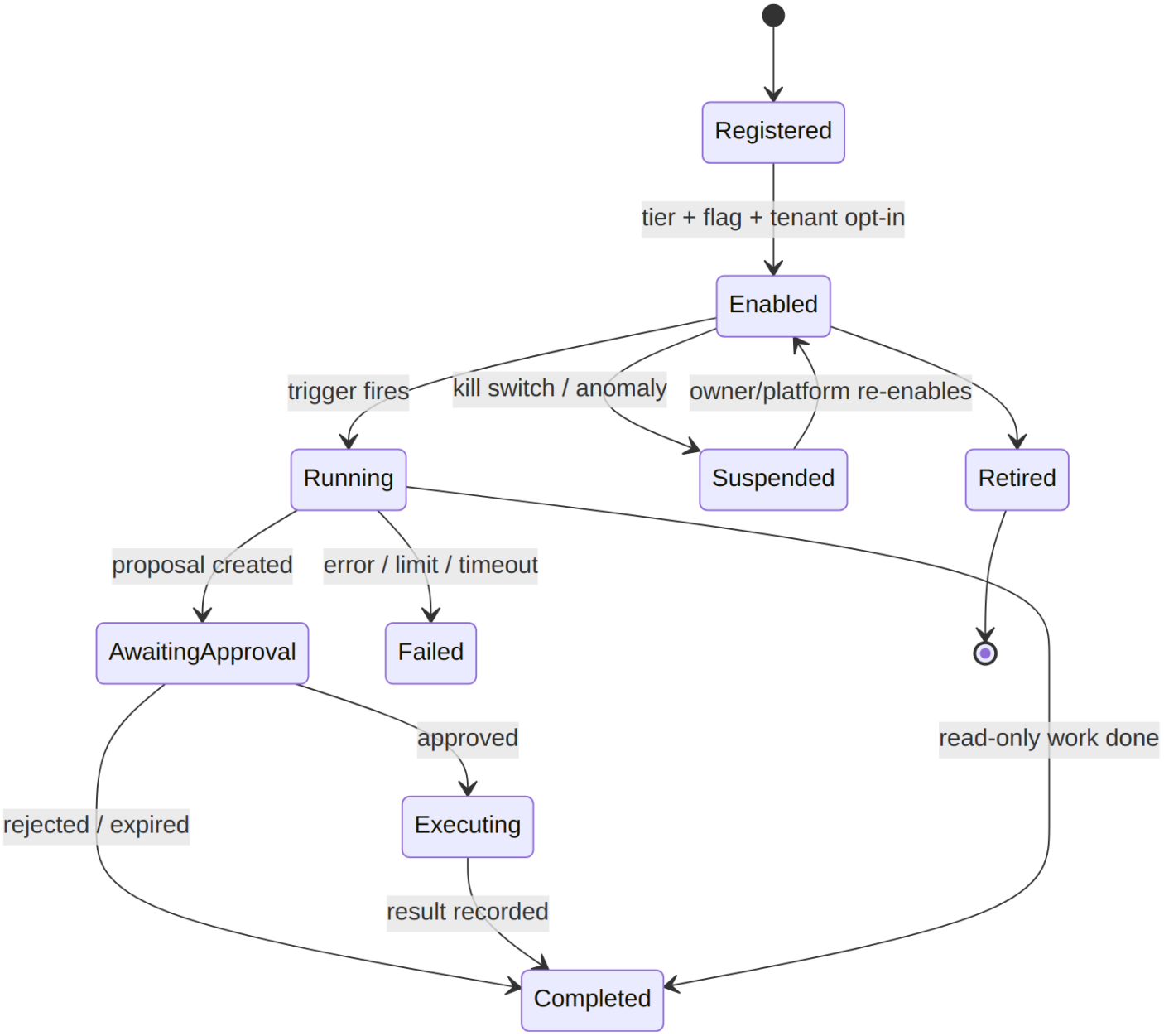
Agents collaborate through **workflow events (outbox)**, not direct agent-to-agent calls (SAD §9).



10.1 Agent Registry

Catalogue of agent types (metadata, not code): agent ID, version, domain, whitelisted tool IDs, allowed triggers, risk class, required approval class, prompt version, phase, enabled-by-tier flag, rate limits, owner. A tenant-level enablement record controls which agents are active for which tenant (BR-AI-11). Adding an agent = registry entry + tool wrappers + prompt + evaluation set; no orchestrator change.

10.2 Agent Lifecycle



A kill switch (per agent, per tenant, platform-wide) is available from Phase 2 **[Proposed]**.

10.3 Task Assignment

- Tasks are created from triggers with an idempotency key (agent + tenant + trigger + window) to avoid duplicates.
- Scheduler selects the registered agent, injects the tenant-scoped service principal, and applies rate/cost limits.
- Priority: user-initiated > event-driven > scheduled; per-tenant concurrency cap.
- Task fan-out (e.g. Studio Insights requesting analysis) is by publishing events, consumed by the target agent's trigger rule.

10.4 Agent Communication

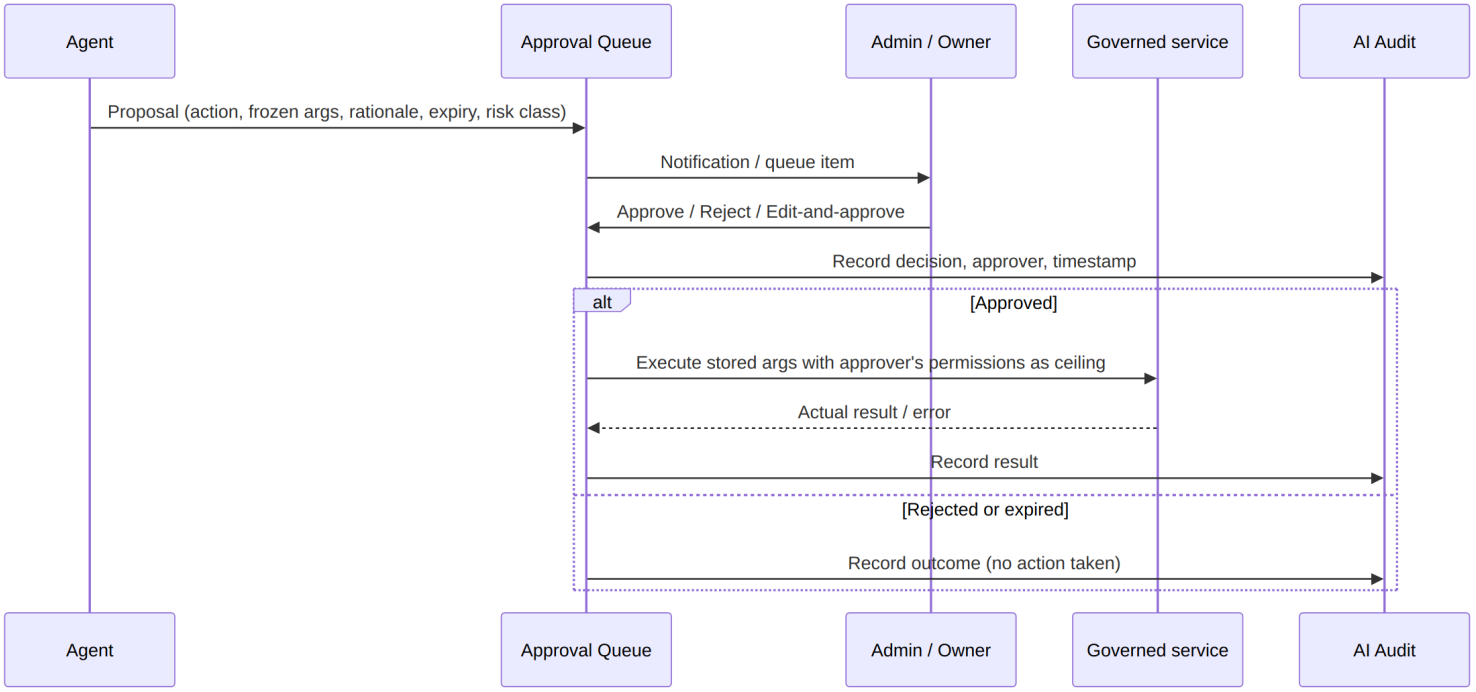
- **Between agents:** outbox events with the standard envelope (event ID, type, tenant, occurred-at, subject reference, correlation ID); consumers idempotent. Example: `booking.cancelled` → Operations Agent updates capacity insight; CRM Agent proposes a re-engagement follow-up.
- **Agent ↔ human:** approval queue items and alerts.
- **Agent ↔ platform:** tool registry only.
- Cross-agent chains carry a **chain depth limit** to prevent loops **[Proposed]**.

10.5 Execution Tracking

Each run records: run ID, agent, version, tenant, trigger, correlation ID, status, tool calls (linked to AI Audit), tokens/cost, duration, proposals created, outcomes. Visible in an Owner/Admin activity view. Alerts on failure rate, cost spikes, and runaway tool loops (max

steps per run).

10.6 Approval Workflow

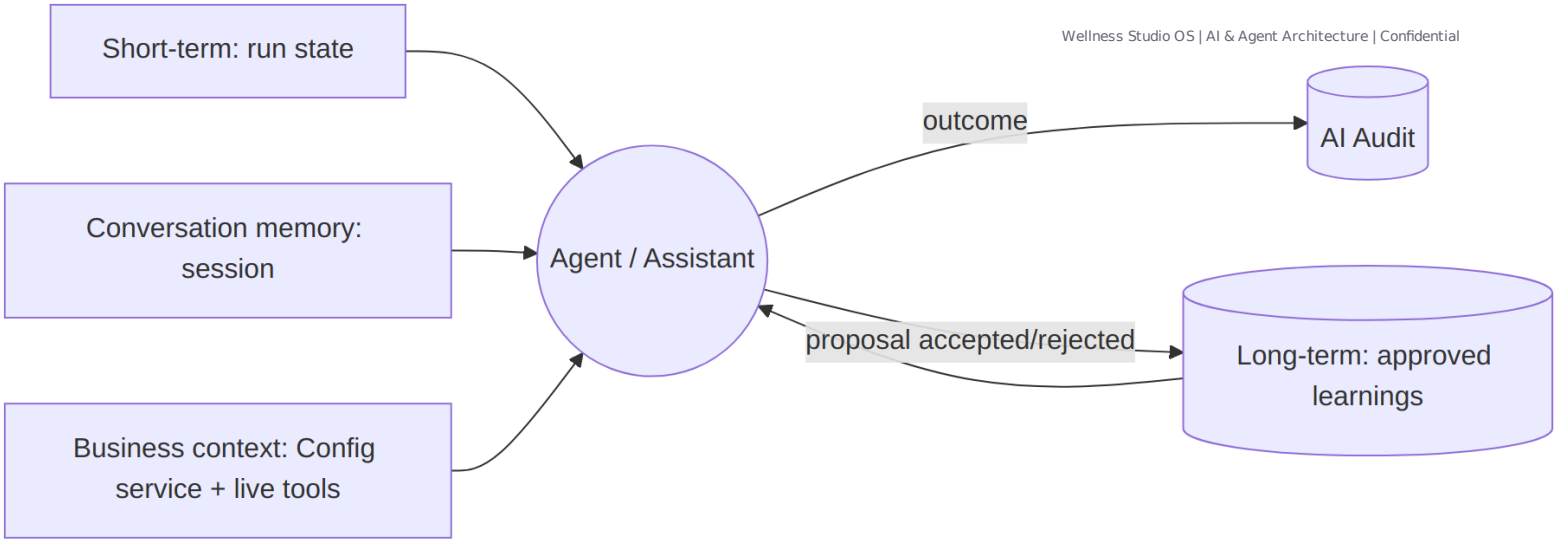


Rules: only stored arguments execute; edits create a new proposal version; approver must hold the permission for the underlying action; expired proposals cannot be executed; approval classes: *none* (alerts), *single* (Admin), *owner-only* (spend, publication, pricing).

11. Agent Memory Architecture

11.1 Memory Types

Memory	Scope	Contents	Storage	Retention
Short-term (working)	One task/run	Plan, intermediate tool results, scratch state	In-process/run store; **[Proposed]** run record	Discarded at run end; only outcome kept in audit
Conversation	One chat session/user	Turns, resolved entities, pending-action reference	Ephemeral session store (Phase 1: in-memory/short-TTL App DB); PII not retained beyond session (MS BR-AI-10)	Short TTL; not business data (BR-AI-07)
Long-term	Tenant, cross-run	Approved learnings: accepted/rejected proposal history, approved knowledge entries, evaluation outcomes	App DB, tenant-scoped, human-approved before use	Per tenant policy; deletable
Business context	Tenant	Studio profile, class types, plans, policies, tone/brand voice, tier/flags, vertical	Config service (App DB/ERP), read via Config service	Live; not copied



11.2 Storage Strategy

- Everything tenant-scoped with RLS; no cross-tenant memory.
- Memory holds **references (IDs)**, not copies of business facts; facts are re-fetched live (avoids stale or second-source truth).
- Member PII excluded from long-term memory; if personalization needs history, it is derived on demand from attendance counts.
- Only content promoted through a governed, human-approved path enters long-term memory (no silent self-learning).

11.3 Retrieval Strategy

1. Always load business context (small, cached by Config service).
2. Load conversation memory for the current session only.
3. Load long-term entries relevant to the task type and tenant (e.g. "this studio rejects Thursday cancellations proposals") by key/tag lookup; semantic retrieval **[Proposed, Phase 3]**.
4. Cap tokens per memory type; oldest/least relevant truncated first; the model receives summaries rather than raw histories.

11.4 Governance

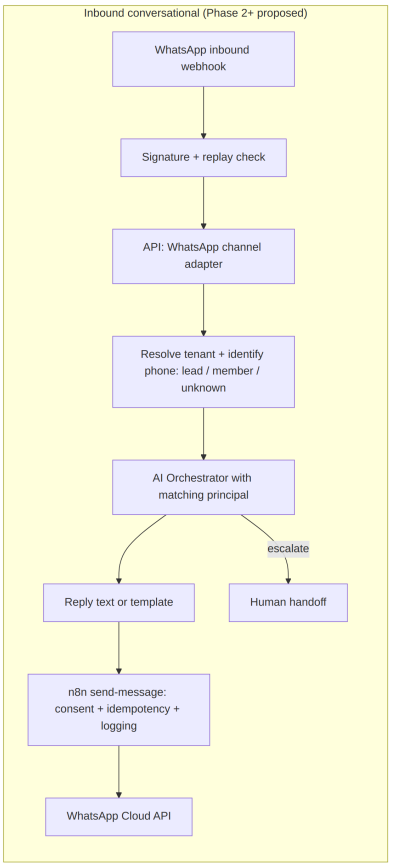
Members' right-to-deletion propagates to memory (legal review pending); memory reads are logged in audit for agents; prompt-injection resistant: memory content is data, never instructions (Section 13.3).

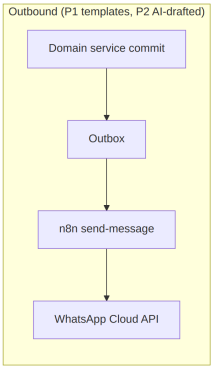
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12. AI & WhatsApp Integration

Baseline (sources): WhatsApp is delivered via n8n and the WhatsApp Cloud API; Phase 1 runs in test/mock mode with template messages only. AI does not read or write WhatsApp directly. Everything below beyond template notifications is **[Proposed] Phase 2+**, and needs production credentials and consent review first.

12.1 Integration Pattern





Design rule: the AI never holds WhatsApp credentials. Replies are released only through the same send-message sub-workflow that enforces templates, consent, idempotency and delivery logging.

12.2 Lead Conversations

- Prospect messages → resolved as **visitor principal** (Enquiry Assistant tools only) → grounded answers → lead capture with consent → `lead.created`.
- Existing lead → CRM Agent (Phase 2) drafts follow-ups; **Admin approves before send**; sequences respect consent/template rules (BR-CRM-04).
- Phase 1: only the templated lead confirmation is sent.

12.3 Booking Assistance

- Recognised member (verified phone) asks "Any Zumba tomorrow evening?" → Booking Assistant tools → shortlist with deep links.
- **Phase 2:** in-chat booking uses propose → confirm (the member replies "Confirm" to a stored proposal) through the standard booking service; final eligibility/capacity enforced in ERP. Phase 1 = link handoff only.
- Cancellation-cutoff and waitlist messaging remain template-driven.

12.4 Follow-Up Messages

- Transactional (confirmation, reminders, renewal) stay **deterministic templates** on n8n schedules; AI does not alter them.

- AI-drafted, non-transactional follow-ups (CRM/Marketing agents): draft → approval queue → approved template or 24-hour session-window message → send-message sub-workflow. Marketing-type messages require opt-in; opt-outs honoured.

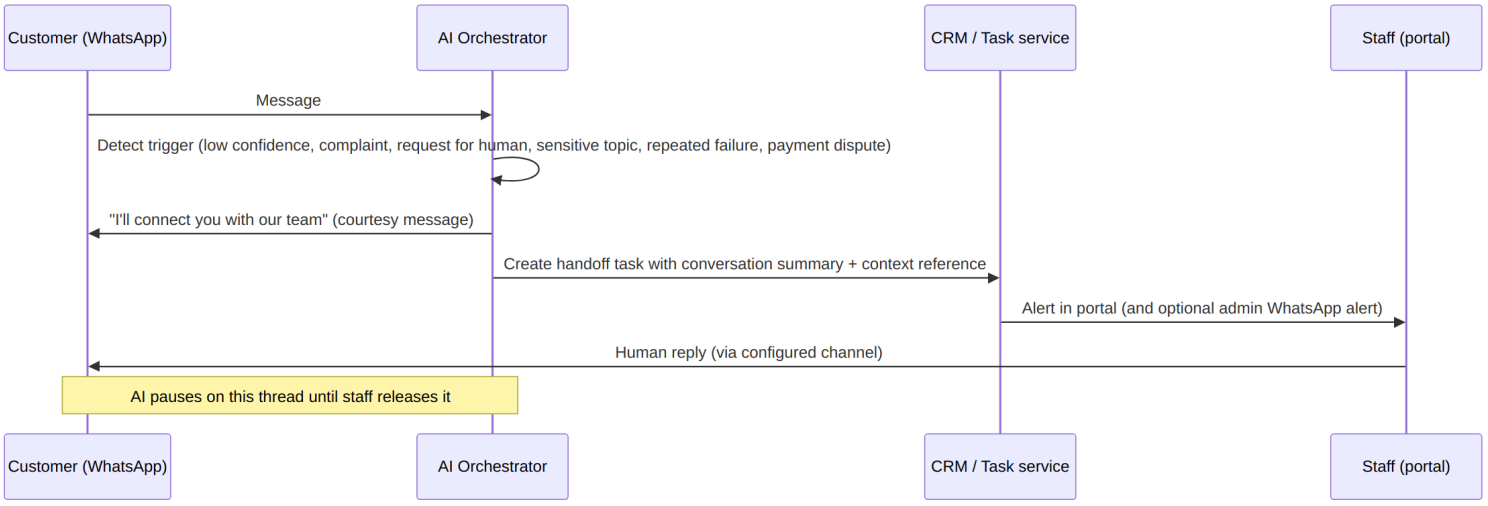
12.5 Member Support

- Identified member, own-data intents only: membership status/expiry, upcoming bookings, class availability, payment status (read-only, statuses exactly as recorded; pending never presented as paid).
- Any action (pay, renew, cancel) links to the authenticated portal; no financial or destructive action from chat in Phase 2.
- Identity: WhatsApp phone match to a verified member phone; sensitive data requests may require OTP step-up **[Proposed]**.

12.6 Trigger Flow

Trigger	Path
Inbound message	Webhook → verify → resolve tenant/identity → Orchestrator → reply via n8n
Business event	Outbox → n8n → template send (no AI)
Agent draft approved	Approval queue → n8n send-message
Schedule	n8n schedule → API "what is due" → template send

12.7 Escalation Flow and Human Handoff



Handoff rules: AI stops replying on the thread; summary contains only necessary context; SLA target and unassigned-handoff alerts are tenant-configurable **[Proposed]**; outside business hours the customer receives the tenant's configured hours message.

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13. AI Security & Governance

13.1 Access Controls

- Evaluation chain: identity → role → permission → tenant/studio/branch → resource ownership (SAD §12.2), applied to tool listing **and** re-checked at execution.
- Principals: staff (user permissions), member (own records), visitor (public-read + create-lead), agent service principal (one tenant, one domain toolset, user ceiling).
- Front Desk: read tools only when enabled by Owner; Trainers: no Assistant access; Trainers get aggregate feedback only.
- Tier/flag gating server-side (BR-AI-11); cost/token quotas per tenant and tier.
- ERP access only via restricted integration identity from the API layer; AI holds none.

13.2 Data Privacy

- Prompts contain the minimum necessary; redaction of PII before prompts and logs (NFR-SEC-05).
- Excluded from all AI: health notes, payment details, emergency contacts, credentials.

- Provider contracts must specify no training on tenant data and retention terms **[Open]**; provider region/data-sovereignty options preserved by model-agnostic gateway and open-source/local model evaluation in Phase 2 (PRD 4.5).
- Consent, purpose limitation, retention and deletion aligned with Indian data protection law; legal review pending (NFR-CMP-01).
- Encryption in transit and at rest.

13.3 Prompt Security

Threat	Control
Prompt injection via user text, FAQ text, feedback comments, lead names	Treat all retrieved/user content as **data**; delimited and labelled in prompts; tool arguments schema-validated; permissions enforced outside the model
Tool abuse / privilege escalation	Registry filtering; server-side re-check; no tool registered for prohibited actions (e.g. no `bookSession` in Phase 1)
Argument drift after confirmation	Pending Action Store holds frozen arguments; only stored args execute (BR-AI-09)
Cross-tenant leakage	Tenant from host/JWT, never from prompt; tenant-filtered retrieval; no shared index
Data exfiltration via output	Output validation; minimal tool return fields; no secrets in context
Jailbreak to give health/financial advice	Hard scope refusal; evaluation set includes adversarial cases
Abuse / cost	Rate limits (stricter for public and AI endpoints), bot protection, per-tenant quotas, max steps per run
System prompt leakage	Prompts hold no secrets; versioned in repo

13.4 Tenant Isolation

Tool calls run under the invoking tenant; RLS on AI tables (audit, pending actions, conversations, knowledge, vectors); ERP tenant permission hooks; per-tenant credentials and templates; cross-tenant tests in CI include AI tools; cross-tenant attempts audited and alerted (SAD §6.1, §12.3).

13.5 AI Audit Logging

Aspect	Design
What	Every model-initiated tool call (reads and writes), approvals/rejections, agent runs, refusals and escalations
Record	Tenant, actor, actor type (**human / automation / AI**), request, tool, arguments, decision, approval, result, prompt version, model/provider identifier, correlation ID, timestamp
Integrity	Append-only; app role cannot update or delete; written in the same transaction as outbox event where applicable
Access	Owner full, Admin limited, read-only (export Phase 2)
Evaluation	Harness reads audit for false-success detection; incidents reviewed

13.6 Human Oversight

- Confirmation for transactional actions; approval queue for agent proposals; kill switches; activity views for Owner.
- Quality gates in CI: curated query set, Create Member scenarios (missing fields, duplicate, permission denied, API failure), zero-false-success assertion, tenant-isolation and scope-refusal tests. Results recorded per release.
- Incident process: false-success or cross-tenant event = Sev-1; feature disabled by flag until fixed.
- Periodic review of prompts, tool registry and agent proposal acceptance rates by named owners.

14. MVP AI Scope

14.1 AI Features

Priority	Item	Note
P0	LLM Gateway (single provider) with timeouts, one retry	Day 2 skeleton
P0	AI Orchestrator, Tool Registry (permission/tier filtered), Guardrails, Pending Action Store, AI Audit	Days 1–4

Priority	Item	Note
P0	Admin Assistant: query 1 (expiring memberships) and query 2 (low-attendance classes), matching dashboard figures	Day 3
P0	Admin Assistant: Create Member with missing-field handling, confirmation, true result	Day 4
P0	Scope refusal, tier/feature-flag gating, evaluation harness for the above	Days 4–5
P1	Additional read tools T-05/T-06 (member count, dashboard summary)	Cheap add-on if time
P1	Enquiry Assistant (narrow)	Day 5; cut order #3
P1	Feedback Assistant (classify + weekly summary)	Day 5; cut order #2
P1	Recommendation Assistant (rules-constrained)	Day 5; cut order #1
P1	Booking Assistant (shortlist, no booking)	Day 5; cut order #1
P2	Conversational booking, personalised recommendations, admin insights, additional Assistant actions with approval queue, feedback trend dashboard, RAG/vector retrieval, open-source model evaluation	Phase 2

14.2 Agent Features

Priority	Item
P0	None built. Only the **agent-ready foundations**: shared registry, service-layer tools, audit with actor type "AI", pending-action store, event outbox
P1	Agent Registry schema and approval-queue **design** reviewed (no build), so Phase 2 needs no redesign
P2	Agent Runtime, Approval Queue UI, CRM Agent, Operations Agent (read and draft), basic Session Summaries; execution tracking; kill switch. Later: Marketing, Advertising, Growth, Studio Insights (Phase 3); Session Summary Agent (Future)

14.3 WhatsApp AI Features

Priority	Item
P0	None AI-driven. Template notifications run via n8n in mock mode (non-AI, platform scope)
P1	Enquiry Assistant lead → `lead.created` → templated WhatsApp lead confirmation (mock)
P2	Inbound AI conversations (lead + member), WhatsApp booking shortlist, AI-drafted follow-ups with approval, human handoff tasks, in-chat booking with confirmation, production credentials and opt-in management

14.4 Now vs Later

Capability	Now (Phase 1)	Later
Actions by AI	Create Member with confirmation	More actions with approval queue (P2)
Autonomy	None	Draft/approve agents (P2–P3); bounded auto-execution only after evidence (**[Open]**)
Retrieval	Tools + small approved FAQ	Semantic RAG (P2)
Messaging	Templates only	AI conversations and drafts (P2+)
Model	One commercial provider via gateway	Evaluate open-source/local (P2)

Cut order (PRD §16.1): recommendations/booking suggestions → feedback → enquiry → paid-event payment → Create Member → read queries. Core operational modules, permissions, tenant isolation and audit are never cut for AI.

15. Future AI Roadmap

Phase	Features	Benefits	Dependencies
1 — MVP AI (5-day delivery)	Admin Assistant (two reads + Create Member); narrow Enquiry, Booking, Recommendation, Feedback assistants; gateway, orchestrator, registry, guardrails, pending actions, audit, evaluation harness	Proof of safe governed AI; faster admin; lead capture; agent-ready foundation	Domain services and reports live (Days 1–3); audit; provider chosen; feature flags; approved FAQ list
2 — Advanced Recommendations (4–8 weeks after Phase 1)	Conversational booking with confirmation; personalised recommendations from history; more Assistant queries/actions + approval queue; admin insights; RAG over approved knowledge; feedback trend dashboard; retention at-risk lists; basic session summaries; open-source/local model evaluation; CRM and Operations Agents (read/draft) begin	Higher booking conversion and retention; less manual monitoring; measurable acceptance rates	Production WhatsApp and payment credentials; consent/preferences (in-app centre); approval-queue UI; data volume from pilot; Phase 1 evaluation baselines; legal review
3 — Semi-Autonomous Agents	Agent Runtime with registry, tracking, kill switch; CRM and Operations Agents production-grade; Marketing and Advertising Agents (approval for every send, publish, spend); inbound WhatsApp AI with handoff; AI-assisted send after approval	Reduced lead leakage; campaign productivity; proactive alerts	Approval workflow proven in Phase 2; consent/opt-in enforcement; ad platform integration with no publish credentials for agents; cost controls
4 — Multi-Agent Intelligence	Growth Optimization and Studio Insights Agents; cross-agent workflows through events; churn/demand forecasts; Session Summary Agent; optional custom predictive models; **[Open]** bounded autonomy for low-risk reversible actions	Revenue and utilisation optimisation; early-warning intelligence; owner-level briefings	Sufficient historical data; stable event taxonomy; agent evaluation suites; governance review before any autonomy increase. Research track (movement/posture intelligence) stays outside this roadmap

Phase numbering note: PRD phases place Marketing, Advertising, Growth and Studio Insights in Phase 3 and Session Summary in Future. This roadmap splits Phase 3 into "semi-autonomous" (marketing/advertising with approval) and Phase 4 "multi-agent intelligence" (growth/insights) for delivery sequencing, which is a **[Proposed]** refinement, not a change to PRD scope.

Appendix A — Tool Inventory (from Module Specification)

ID	Tool	Assistant	Type
T-01	getExpiringMemberships	Admin	Read
T-02	getLowAttendanceClasses	Admin	Read
T-03	extractCreateMemberFields	Admin	Internal
T-04	createMember	Admin	Write, confirm
T-05	getMemberCount	Admin	Read
T-06	getDashboardSummary	Admin	Read
T-07	getPublicEventList	Enquiry	Read
T-08	getPublicClassSchedule	Enquiry	Read
T-09	getActivePlansPublic	Enquiry	Read
T-10	captureLead	Enquiry	Write (lead only)
T-11	getEligibleSchedule	Booking	Read
T-12	getAvailability	Booking	Read
T-13	getMemberPlanEntitlement	Booking	Read
T-14	getMemberGoalAndLevel	Recommendation	Read
T-15	getMemberAttendanceSummary	Recommendation	Read
T-16	getEligibleSessionsForRecommendation	Recommendation	Read
T-17	classifyFeedback	Feedback	Internal

ID	Tool	Assistant	Type
T-18	storeFeedbackClassification	Feedback	Write
T-19	getFeedbackSummary	Feedback	Read

Appendix B — Open Decisions for This Document

1. Confirm beta AI provider and data-privacy terms for member data (PRD Appendix C).
2. Confirm whether inbound conversational WhatsApp AI is wanted in Phase 2 (this document proposes it).
3. Confirm vector approach (PostgreSQL vector extension) and whether it is needed before the FAQ corpus outgrows keyword search.
4. Confirm approval-queue ownership and approval classes (single vs owner-only).
5. Confirm retention periods for conversation memory, feedback and AI audit (legal review).
6. Confirm the feedback collection channel (portal vs post-class WhatsApp link) and whether Trainers see aggregates only.